# H2 MATHEMATICS (9758)

## Complete Formula Sheet

Singapore-Cambridge GCE A-Levels | 2025 / 2026

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### How To Use This Sheet

This formula sheet covers all major topics in the H2 Mathematics (9758) syllabus. Formulas not provided in the exam are marked with **[MEMORISE]**. Items given in the MF26 formula booklet are marked **[MF26]**.

## 1. PURE MATHEMATICS

### 1.1 Indices & Surds

| Identity / Rule | Formula | Notes |
|---|---|---|
| **Product rule** | $a^m \times a^n = a^{(m+n)}$ | *[MEMORISE]* |
| **Quotient rule** | $a^m \div a^n = a^{(m-n)}$ | *[MEMORISE]* |
| **Power rule** | $(a^m)^n = a^{(mn)}$ | *[MEMORISE]* |
| **Zero exponent** | $a^0 = 1 \ (a \neq 0)$ | *[MEMORISE]* |
| **Negative exponent** | $a^{(-n)} = 1 / a^n$ | *[MEMORISE]* |
| **Fractional exponent** | $a^{(m/n)} = \sqrt[n]{a^m}$ | *[MEMORISE]* |
| **Surd rationalisation** | $(a+\sqrt{b})(a-\sqrt{b}) = a^2-b$ | *Multiply by conjugate* |

### 1.2 Quadratics

| Identity / Rule | Formula | Notes |
|---|---|---|
| **Quadratic formula** | $x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$ | *[MEMORISE]* |
| **Discriminant** | $\Delta = b^2 - 4ac$ | *$\Delta>0$: 2 real roots* |
| **Sum of roots** | $\alpha + \beta = -b/a$ | *[MEMORISE]* |
| **Product of roots** | $\alpha\beta = c/a$ | *[MEMORISE]* |
| **Completing the square** | $ax^2+bx+c = a(x+b/2a)^2+c-b^2/4a$ | *Standard method* |

**Exam Tip:** *For 'always positive' / 'always negative' questions: check the discriminant AND leading coefficient. State both clearly in your working.*

## 1.3 Functions

| Concept | Key Rule / Formula | Notes |
|---|---|---|
| **Composite function** | $(f \circ g)(x) = f(g(x))$ | *Apply g first, then f* |
| **Inverse exists when** | f is one-to-one (injective) | *[MEMORISE]* |
| **Domain of $f^{-1}$** | Range of f | *Always state domain* |
| **Range of $f^{-1}$** | Domain of f | *Always state range* |
| **$f \circ f^{-1}$ = identity** | $f(f^{-1}(x)) = x$ | *Valid on domain of $f^{-1}$* |
| **Even function** | $f(-x) = f(x)$ | *Symmetric about y-axis* |
| **Odd function** | $f(-x) = -f(x)$ | *Rotational symmetry* |

## 1.4 Exponential & Logarithms

| Identity / Rule | Formula | Notes |
|---|---|---|
| **Definition** | y = log_a(x) ⇔ x = a^y | *a > 0, a ≠ 1, x > 0* |
| **Product rule** | log(AB) = log A + log B | *[MEMORISE]* |
| **Quotient rule** | log(A/B) = log A - log B | *[MEMORISE]* |
| **Power rule** | log(A^n) = n log A | *[MEMORISE]* |
| **Change of base** | log_a(b) = ln b / ln a | *[MEMORISE]* |
| **Natural log identity** | ln(e^x) = x and e^(ln x) = x | *[MEMORISE]* |
| **Graph of e^x** | Passes (0,1), asymptote y=0 | *Always show asymptote* |
| **Graph of ln x** | Passes (1,0), asymptote x=0 | *Always show asymptote* |

**Exam Tip:** *ln and $\log_{10}$ are different. In H2 Math, 'log' without a base usually means ln. Always check context.*

## 1.5 Trigonometry

| Identity | Formula | Notes |
|---|---|---|
| **Pythagorean identity** | $\sin^2\theta + \cos^2\theta = 1$ | *[MEMORISE]* |
| **tan identity** | $\tan\theta = \sin\theta / \cos\theta$ | *[MEMORISE]* |
| **sec² identity** | $1 + \tan^2\theta = \sec^2\theta$ | *[MEMORISE]* |
| **cosec² identity** | $1 + \cot^2\theta = \mathrm{cosec}^2\theta$ | *[MEMORISE]* |

| Identity | Formula | Notes |
|---|---|---|
| **Double angle (sin)** | $\sin 2A = 2 \sin A \cos A$ | *[MF26]* |
| **Double angle (cos)** | $\cos 2A = \cos^2 A - \sin^2 A$ | *[MF26]* |
| **cos 2A (alt forms)** | $= 2\cos^2 A - 1 = 1 - 2\sin^2 A$ | *[MF26]* |
| **Double angle (tan)** | $\tan 2A = 2\tan A / (1 - \tan^2 A)$ | *[MF26]* |
| **Addition (sin)** | $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ | *[MF26]* |
| **Addition (cos)** | $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ | *[MF26]* |
| **R-formula** | $a \sin\theta + b \cos\theta = R \sin(\theta + \alpha)$ | *$R=\sqrt{(a^2+b^2)}$, $\tan \alpha=b/a$* |
| **Factor formulae** | $\sin P + \sin Q = 2\sin((P+Q)/2)\cos((P-Q)/2)$ | *[MF26]* |

**Exam Tip:** *R-formula: Always find $R > 0$ first, then find $\alpha$ in the correct quadrant. State the range of $(\theta + \alpha)$ before solving.*

## 1.6 Differentiation

| Function | Derivative d/dx | Notes |
|---|---|---|
| **x^n** | n x^(n-1) | *[MEMORISE]* |
| **e^x** | e^x | *[MEMORISE]* |
| **e^(f(x))** | f'(x) · e^(f(x)) | *Chain rule* |
| **ln x** | 1/x | *[MEMORISE]* |
| **ln f(x)** | f'(x) / f(x) | *Chain rule* |
| **sin x** | cos x | *[MEMORISE]* |
| **cos x** | -sin x | *[MEMORISE]* |
| **tan x** | $\sec^2 x$ | *[MEMORISE]* |
| **$\sin^{-1} x$** | $1 / \sqrt{(1-x^2)}$ | *[MF26]* |
| **$\cos^{-1} x$** | $-1 / \sqrt{(1-x^2)}$ | *[MF26]* |
| **$\tan^{-1} x$** | $1 / (1+x^2)$ | *[MF26]* |
| **Product rule** | d/dx(uv) = u'v + uv' | *[MEMORISE]* |
| **Quotient rule** | $d/dx(u/v) = (u'v - uv') / v^2$ | *[MEMORISE]* |
| **Chain rule** | dy/dx = dy/du · du/dx | *[MEMORISE]* |
| **Implicit diff.** | Differentiate both sides w.r.t. x | *Use d/dx(y) = dy/dx* |
| **Parametric diff.** | dy/dx = (dy/dt) / (dx/dt) | *[MEMORISE]* |

**Exam Tip:** *For connected rates: $dA/dt = dA/dr \times dr/dt$. Always identify which rate is given and which is required before starting.*

## 1.7 Integration

| Function f(x) | Integral ∫f(x) dx | Notes |
|---|---|---|
| **$x^n$ ($n \neq -1$)** | $x^{n+1} / (n+1) + C$ | *[MEMORISE]* |
| **$1/x$** | $\ln\lvert x\rvert + C$ | *Note absolute value* |
| **$e^x$** | $e^x + C$ | *[MEMORISE]* |
| **$e^{(ax+b)}$** | $e^{(ax+b)} / a + C$ | *[MEMORISE]* |
| **sin x** | $-\cos x + C$ | *[MEMORISE]* |
| **cos x** | $\sin x + C$ | *[MEMORISE]* |
| **sin(ax+b)** | $-\cos(ax+b) / a + C$ | *[MEMORISE]* |
| **cos(ax+b)** | $\sin(ax+b) / a + C$ | *[MEMORISE]* |
| **$\sec^2 x$** | $\tan x + C$ | *[MEMORISE]* |
| **$1/(a^2+x^2)$** | $(1/a) \tan^{-1}(x/a) + C$ | *[MF26]* |
| **$1/\sqrt{(a^2-x^2)}$** | $\sin^{-1}(x/a) + C$ | *[MF26]* |
| **$1/(x^2-a^2)$** | $(1/2a) \ln\lvert(x-a)/(x+a)\rvert + C$ | *[MF26]* |
| **Integration by parts** | $\int u\,dv = uv - \int v\,du$ | *[MEMORISE]* |

**Exam Tip:** *LIATE rule for integration by parts: Logarithmic > Inverse trig > Algebraic > Trigonometric > Exponential. Pick the first type as u.*

## 1.8 Maclaurin Series

| Series | Expansion | Notes |
|---|---|---|
| **General Maclaurin** | $f(x) = f(0) + f'(0)x + f''(0)x^2/2! + \ldots$ | *[MF26] — know structure* |
| **$e^x$** | $1 + x + x^2/2! + x^3/3! + \ldots$ | *[MF26]* |
| **sin x** | $x - x^3/3! + x^5/5! - \ldots$ | *[MF26]* |
| **cos x** | $1 - x^2/2! + x^4/4! - \ldots$ | *[MF26]* |
| **ln(1+x)** | $x - x^2/2 + x^3/3 - \ldots \quad \lvert x\rvert<1$ | *[MF26]* |
| **$(1+x)^n$** | $1 + nx + n(n-1)x^2/2! + \ldots \quad \lvert x\rvert<1$ | *[MF26]* |

**Exam Tip:** *Lost marks source: Forgetting to state the range of validity $\lvert x\rvert < 1$ for ln(1+x) and $(1+x)^n$. Always state it.*

## 1.9 Complex Numbers

| Concept | Formula | Notes |
|---|---|---|
| **Definition** | $z = x + iy$ where $i^2 = -1$ | *[MEMORISE]* |

| Concept | Formula | Notes |
|---|---|---|
| Modulus | $\|z\| = \sqrt{x^2 + y^2}$ | *[MEMORISE]* |
| Argument | $\arg(z) = \theta = \tan^{-1}(y/x)$ | *Check quadrant!* |
| Polar form | $z = r(\cos\theta + i\sin\theta)$ | *$r = \|z\|$* |
| Euler form | $z = re^{i\theta}$ | *[MEMORISE]* |
| Conjugate | $z^* = x - iy$ | *$z \times z^* = \|z\|^2$* |
| De Moivre's Theorem | $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ | *[MF26]* |
| Multiplication (mod) | $\|z_1z_2\| = \|z_1\|\|z_2\|$ | *[MEMORISE]* |
| Multiplication (arg) | $\arg(z_1z_2) = \arg(z_1) + \arg(z_2)$ | *[MEMORISE]* |
| Locus $\|z - a\| = r$ | Circle, centre a, radius r | *Sketch on Argand diagram* |
| Locus $\arg(z - a) = \theta$ | Half-line from point a, angle θ | *Exclude the point a* |

**Exam Tip:** *$\arg(z)$ must be in $(-\pi, \pi]$. When finding argument in 3rd or 2nd quadrant, do NOT just use $\tan^{-1}(y/x)$ directly — adjust using $\pi$ or $-\pi$.*

## 1.10 Vectors

| Operation | Formula | Notes |
|---|---|---|
| Magnitude | $\|a\| = \sqrt{a_1^2+a_2^2+a_3^2}$ | *[MEMORISE]* |
| Unit vector | $\hat{a} = a / \|a\|$ | *[MEMORISE]* |
| Dot product | $a \cdot b = \|a\|\|b\|\cos\theta$ | *[MEMORISE]* |
| Dot product (component) | $a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$ | *[MEMORISE]* |
| Perpendicular condition | $a \cdot b = 0$ | *[MEMORISE]* |
| Cross product magnitude | $\|a \times b\| = \|a\|\|b\|\sin\theta$ | *[MF26]* |
| Area of triangle | $(1/2)\|a \times b\|$ | *[MEMORISE]* |
| Projection of a onto b | $(a \cdot \hat{b})\hat{b} = (a \cdot b / \|b\|^2)\, b$ | *[MEMORISE]* |
| Vector line equation | $r = a + \lambda b$ | *a: point, b: direction* |
| Cartesian line (3D) | $(x-a_1)/b_1 = (y-a_2)/b_2 = (z-a_3)/b_3$ | *[MEMORISE]* |
| Plane equation | $r \cdot n = a \cdot n$ | *n: normal vector* |
| Distance: point to plane | $\|a \cdot \hat{n} - d\|$ | *$d = a \cdot \hat{n}$ for any point on plane* |
| Angle between lines | $\cos\theta = \|b_1 \cdot b_2\| / (\|b_1\|\|b_2\|)$ | *Use \|...\| for acute angle* |
| Angle: line and plane | $\sin\theta = \|b \cdot n\| / (\|b\|\|n\|)$ | *[MEMORISE]* |
| Angle: between planes | $\cos\theta = \|n_1 \cdot n_2\| / (\|n_1\|\|n_2\|)$ | *[MEMORISE]* |

**Exam Tip:** *3D Vectors is the highest mark-loss topic. Show all working for geometric interpretations — a correct answer with no method gets 0 marks.*

# 2. STATISTICS

## 2.1 Permutations & Combinations

| Concept | Formula | Notes |
|---|---|---|
| **Permutation (no repeat)** | $^nP_r = n! / (n-r)!$ | *Order matters* |
| **Combination** | $^nC_r = n! / (r!(n-r)!)$ | *Order does NOT matter* |
| **Permutation (with repeats)** | $n! / (p!\ q!\ r!\ \ldots)$ | *Divide by repeated items* |
| **Circular permutation** | $(n-1)!$ | *Fix one item's position* |
| **At least one / none** | Use complement: 1 - P(none) | *[MEMORISE] strategy* |

## 2.2 Probability

| Concept | Formula | Notes |
|---|---|---|
| **Basic probability** | P(A) = favourable / total | *All outcomes equally likely* |
| **Complement** | $P(A') = 1 - P(A)$ | *[MEMORISE]* |
| **Addition rule** | $P(A\cup B) = P(A) + P(B) - P(A\cap B)$ | *[MEMORISE]* |
| **Mutually exclusive** | $P(A\cup B) = P(A) + P(B)$ | *$P(A\cap B) = 0$* |
| **Conditional probability** | $P(A\mid B) = P(A\cap B) / P(B)$ | *[MEMORISE]* |
| **Independent events** | $P(A\cap B) = P(A) \times P(B)$ | *[MEMORISE]* |
| **Independence check** | $P(A\mid B) = P(A)$ | *Equivalent condition* |
| **Total probability** | $P(A) = P(A\mid B)P(B) + P(A\mid B')P(B')$ | *[MEMORISE]* |

## 2.3 Probability Distributions

| Distribution | Formula / Notation | Notes |
|---|---|---|
| **Expectation E(X)** | $\Sigma\, x \cdot P(X=x)$ | *Weighted average* |
| **Variance Var(X)** | $E(X^2) - [E(X)]^2$ | *[MEMORISE]* |
| **E(aX+b)** | $a\,E(X) + b$ | *[MEMORISE]* |
| **Var(aX+b)** | $a^2\,\mathrm{Var}(X)$ | *[MEMORISE]* |
| **Binomial X~B(n,p)** | P(X=r) = nCr · p^r · (1-p)^r | *[MF26]* |

| Distribution | Formula / Notation | Notes |
|---|---|---|
| **Binomial mean** | $E(X) = np$ | *[MEMORISE]* |
| **Binomial variance** | $Var(X) = np(1-p)$ | *[MEMORISE]* |
| **Poisson X~Po(λ)** | $P(X=r) = e^{-\lambda} \lambda^r / r!$ | *[MF26]* |
| **Poisson mean & var** | $E(X) = Var(X) = \lambda$ | *[MEMORISE]* |
| **Poisson approximation** | Use when n large, p small, $np = \lambda$ | *$np < 5$ typically* |
| **Normal X~N(μ,σ²)** | Standardise: $Z = (X-\mu)/\sigma$ | *$Z \sim N(0,1)$* |
| **Normal approximation** | $B(n,p) \approx N(np, np(1-p))$ | *n large, p not too extreme* |
| **Continuity correction** | $P(X \geq 5)$ becomes $P(X>4.5)$ | *Required for Normal approx* |
| **CLT (sample mean X̄)** | $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately | *n large, any distribution* |

**Exam Tip:** *Always verify conditions before using an approximation: state them explicitly for full marks. Forgetting continuity correction loses marks.*

## 2.4 Sampling & Hypothesis Testing

| Concept | Formula / Rule | Notes |
|---|---|---|
| **Unbiased est. of μ** | $\bar{x} = \Sigma x / n$ | *[MEMORISE]* |
| **Unbiased est. of σ²** | $s^2 = (1/(n-1))[\Sigma x^2 - n(\bar{x})^2]$ | *[MEMORISE]* |
| **Test statistic (z-test)** | $Z = (\bar{X} - \mu_0) / (\sigma/\sqrt{n})$ | *[MEMORISE]* |
| **p-value approach** | Reject $H_0$ if p-value $< \alpha$ | *[MEMORISE]* |
| **Critical region** | Reject $H_0$ if $\lvert Z\rvert > z\_critical$ | *Two-tail: $\pm z_{(\alpha/2)}$* |
| **1-tail test** | $H_1: \mu > \mu_0$ or $\mu < \mu_0$ | *Critical region one side* |
| **2-tail test** | $H_1: \mu \neq \mu_0$ | *Split α/2 each tail* |

**Exam Tip:** *Always write $H_0$ and $H_1$ clearly. State the test statistic value. Write a conclusion in context — do not just say 'reject $H_0$'.*

## 2.5 Correlation & Regression

| Concept | Formula | Notes |
|---|---|---|
| **PMCC (r)** | $-1 \leq r \leq 1$ | *$r = \pm 1$: perfect linear* |
| **Regression line y on x** | $y = a + bx$ | *Minimises $\Sigma(y-\hat{y})^2$* |
| **Passes through** | $(\bar{x}, \bar{y})$ | *Always true* |
| **Interpolation vs extrap.** | Interpolation more reliable | *Extrapolation unreliable* |
| **Non-linear: transform** | Use ln y or ln x to linearise | *Then apply regression* |

# 3. QUICK EXAM REFERENCE

| Common Mark-Loss Points | Recovery Strategy |
|---|---|
| Forgetting +C in indefinite integrals | Write +C immediately every time |
| Not stating domain/range of inverse | Template: 'Domain of $f^{-1}$ is range of f' |
| arg(z) in wrong quadrant | Always sketch the Argand diagram first |
| Validity range for Maclaurin series | Write $\|x\| < 1$ as your last line always |
| H-test: no conclusion in context | End with: 'evidence that [μ has/hasn't changed]' |
| Missing continuity correction | ±0.5 whenever approximating discrete → continuous |